November 10, 2026

The Editors
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Dear Editors,

On behalf of my co-authors, Hoang-Viet Vu and Phuc-Son Nguyen, I submit the manuscript *Fair vehicle routing with random route durations: ex-post ordered fairness, rank prices and risk-consistent tours* for consideration as a Theory and Methodology paper in the area of Production, Manufacturing, Transportation and Logistics (first keyword: Routing).

Fair vehicle routing balances drivers' workloads with ordered measures of route lengths, and recent exact work requires every route to be a shortest tour of its customers. All of this work assumes deterministic travel times, whereas drivers experience realised durations. Our paper studies fair routing when travel times are random. We separate ex-ante from ex-post fairness and prove that deterministic fair plans systematically overstate the fairness drivers experience, with a bound that shows the gap is largest exactly for balanced plans. We give an exact, sampling-free integral for the expected value of any ordered measure of independent route durations, derive rank prices as its gradient, show that route consistency must be imposed explicitly under uncertainty, and characterise risk-consistent tours as supported Pareto tours. A bilevel local search builds on these results.

Seventeen experiments on two families of CVRPLIB instances show that deterministic fair plans understate realised unfairness by 8 to 40 percent, that optimising realised fairness improves it consistently (up to 7 percent with generous budgets, while using less of the budget), and that it outperforms sample average approximation and a risk-adjusted deterministic model, which turns out to be no better than ignoring uncertainty.

We believe the paper fits EJOR's scope through its methodological contributions to stochastic and fair routing and its practical message for planners: realised fairness should be reported and optimised, and it can be, at a cost of under a millisecond per plan. The manuscript is original, has not been published, and is not under consideration elsewhere; all authors have approved the submission. An Online Supplement contains the complete computational study and per-instance results.

Yours sincerely,

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